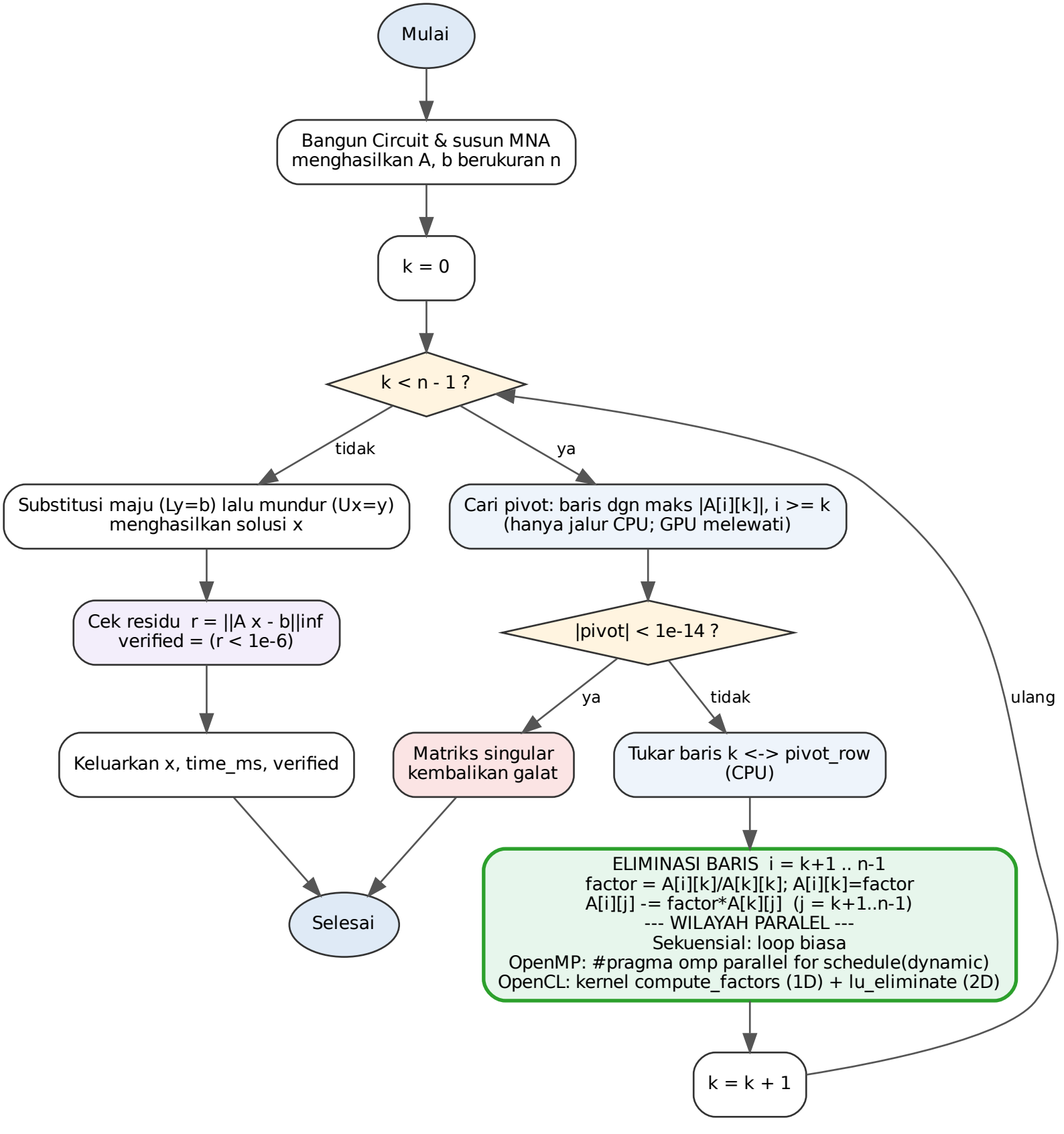
Mulai
Bangun Circuit & susun MNA
menghasilkan A, b berukuran n
k = 0
k < n - 1 ?
tidak
ya
Substitusi maju (Ly=b) lalu mundur (Ux=y)
menghasilkan solusi x
Cari pivot: baris dgn maks |A[i][k]|, i >= k
(hanya jalur CPU; GPU melewati)
Cek residu r = ||A x - b||inf
verified = (r < 1e-6)
|pivot| < 1e-14 ?
ya
tidak
ulang
Keluarkan x, time_ms, verified
Matriks singular
kembalikan galat
Tukar baris k <-> pivot_row
(CPU)
ELIMINASI BARIS i = k+1 .. n-1
factor = A[i][k]/A[k][k]; A[i][k]=factor
A[i][j] -= factor*A[k][j] (j = k+1..n-1)
--- WILAYAH PARALEL ---
Sekuensial: loop biasa
OpenMP: #pragma omp parallel for schedule(dynamic)
OpenCL: kernel compute_factors (1D) + lu_eliminate (2D)
k = k + 1
Selesai